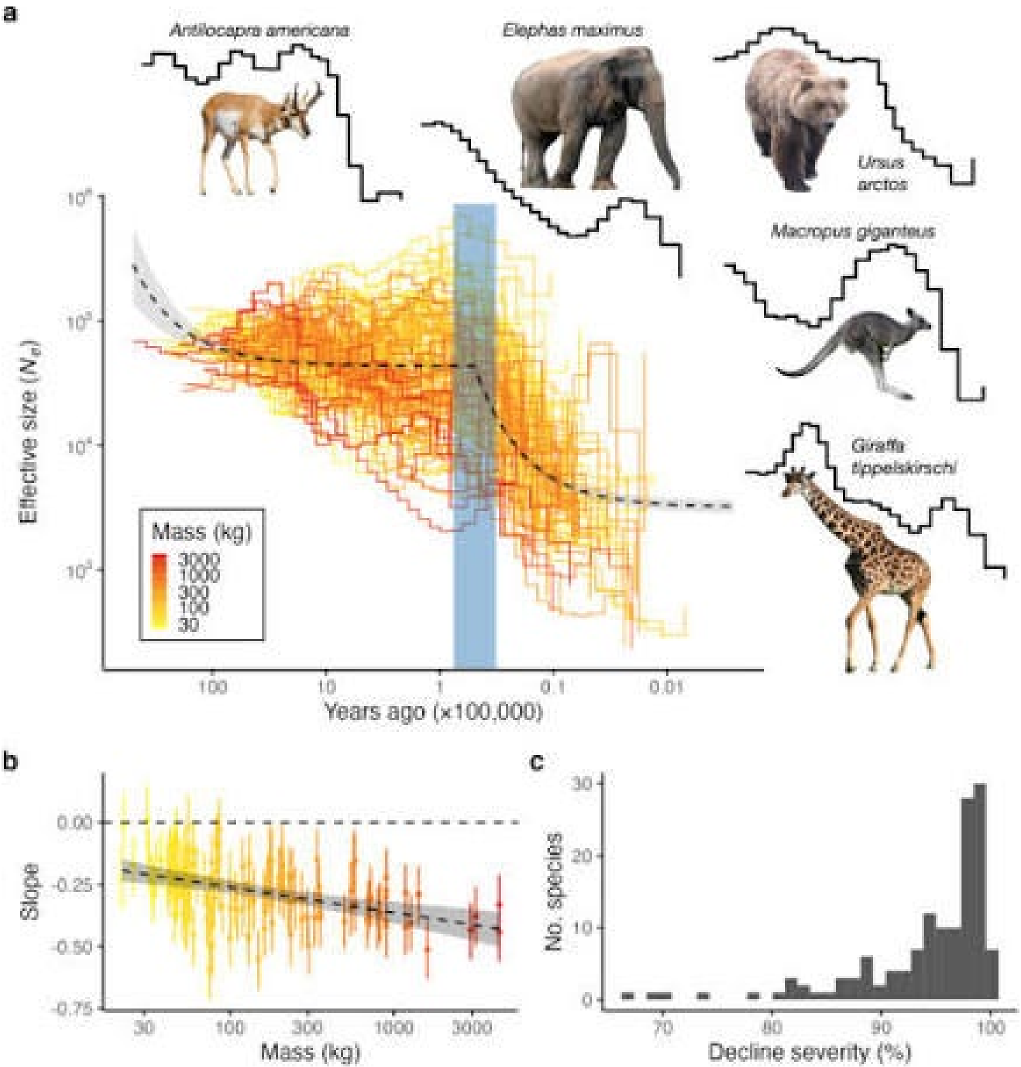
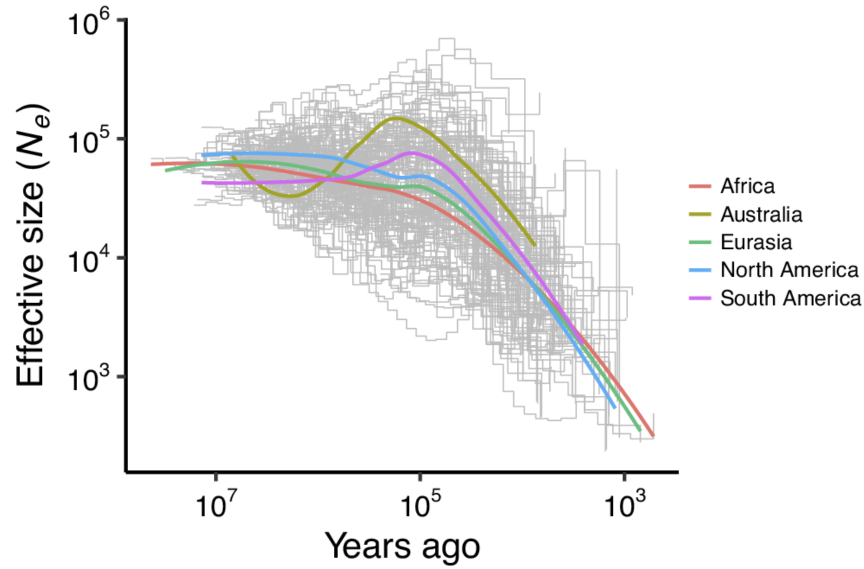
PSMC

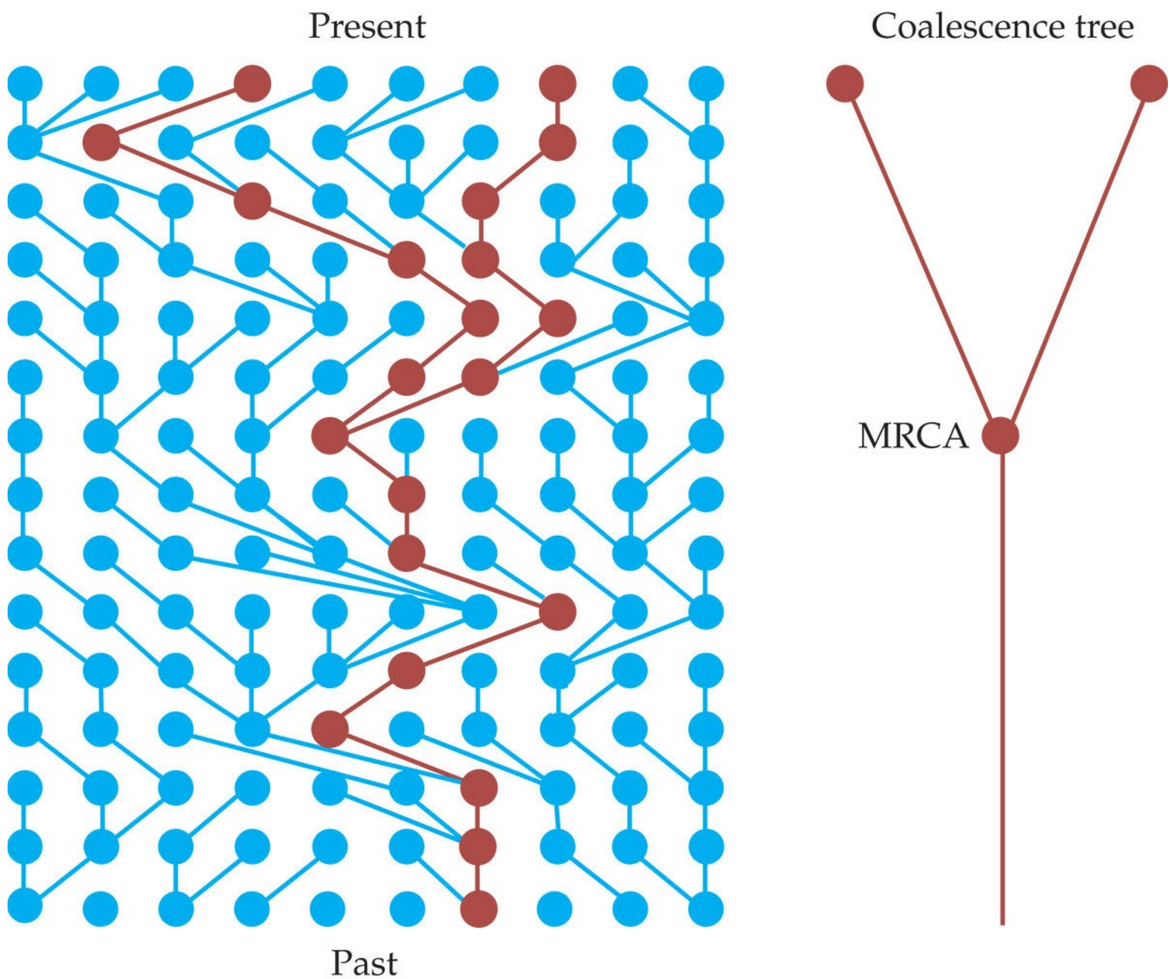
Pairwise Sequential Markovian Coalescent

Mpala workshop, August 2026

Pairwise Sequentially Markovian Coalescent



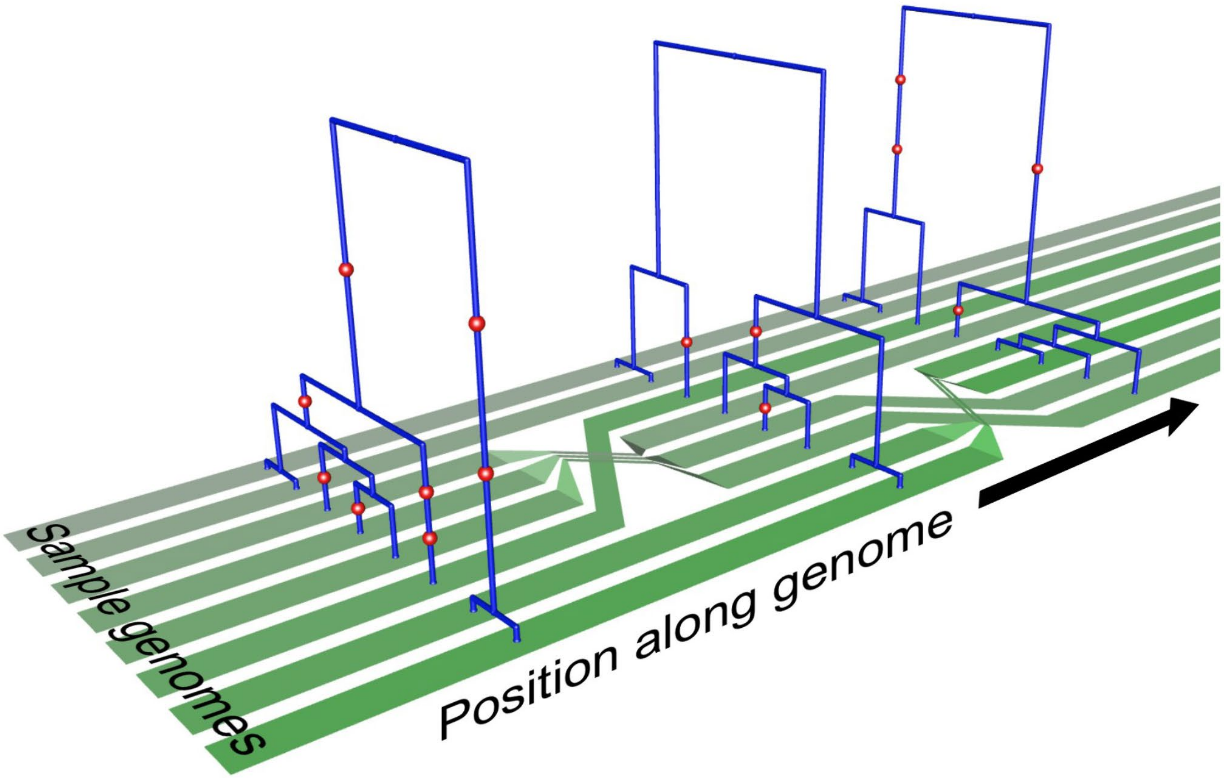




INTRODUCTION TO POPULATION GENETICS, **Figure 3.2**

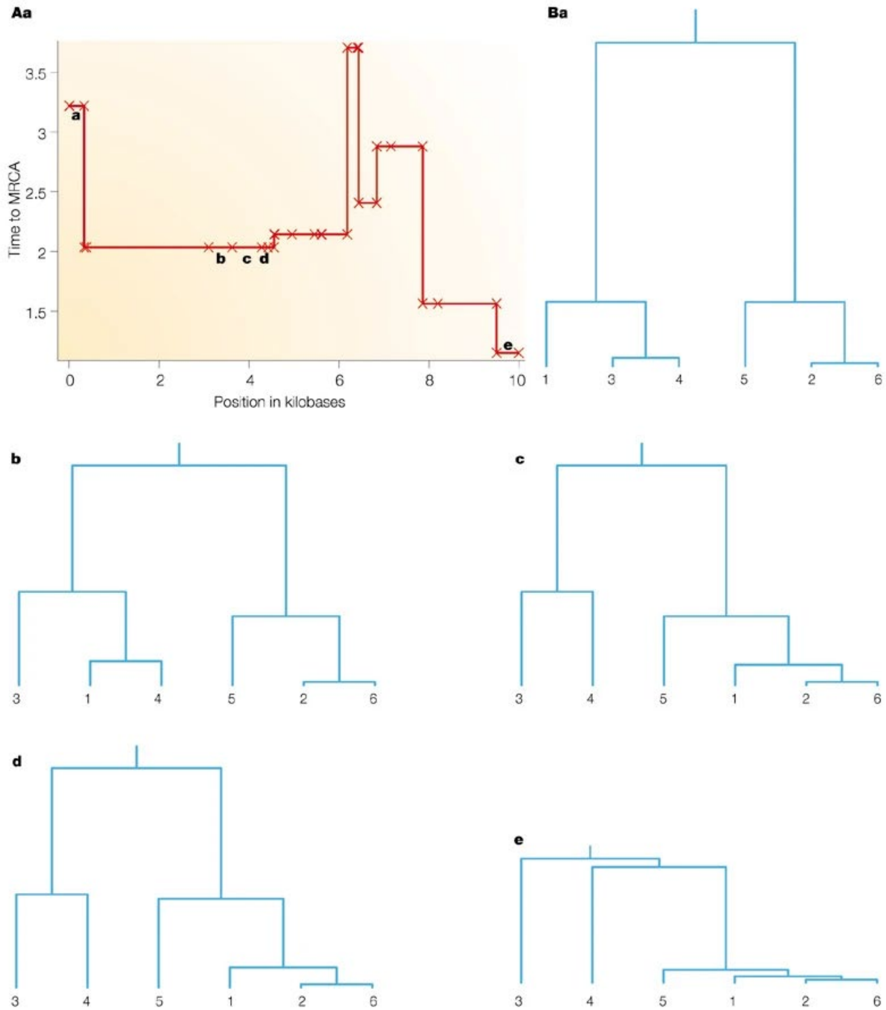
Most Recent Common Ancestor **2*N* generations ago**.

Remember what we said about recombination?



Yan Wong, 2020

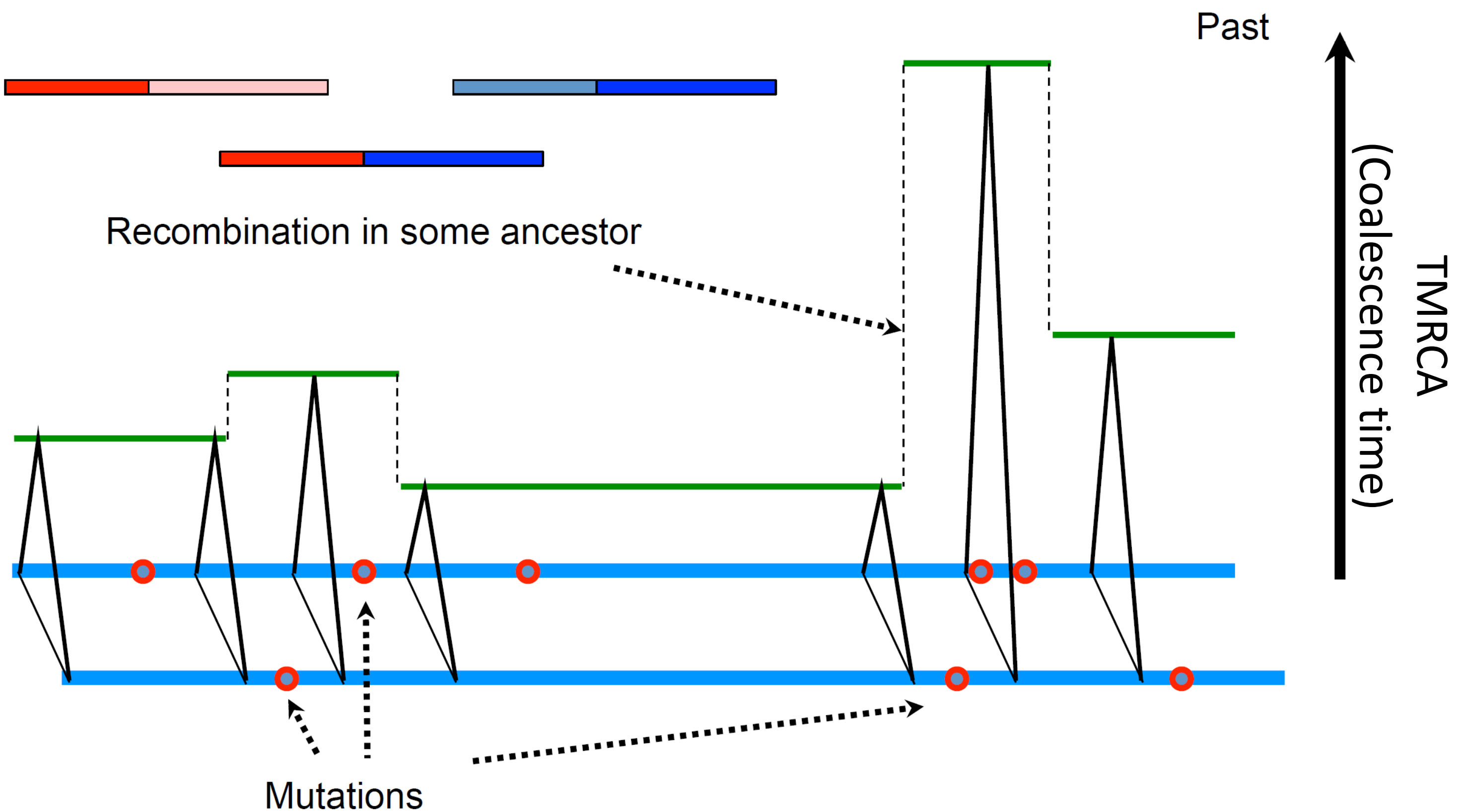
TMRCA changes along the genome



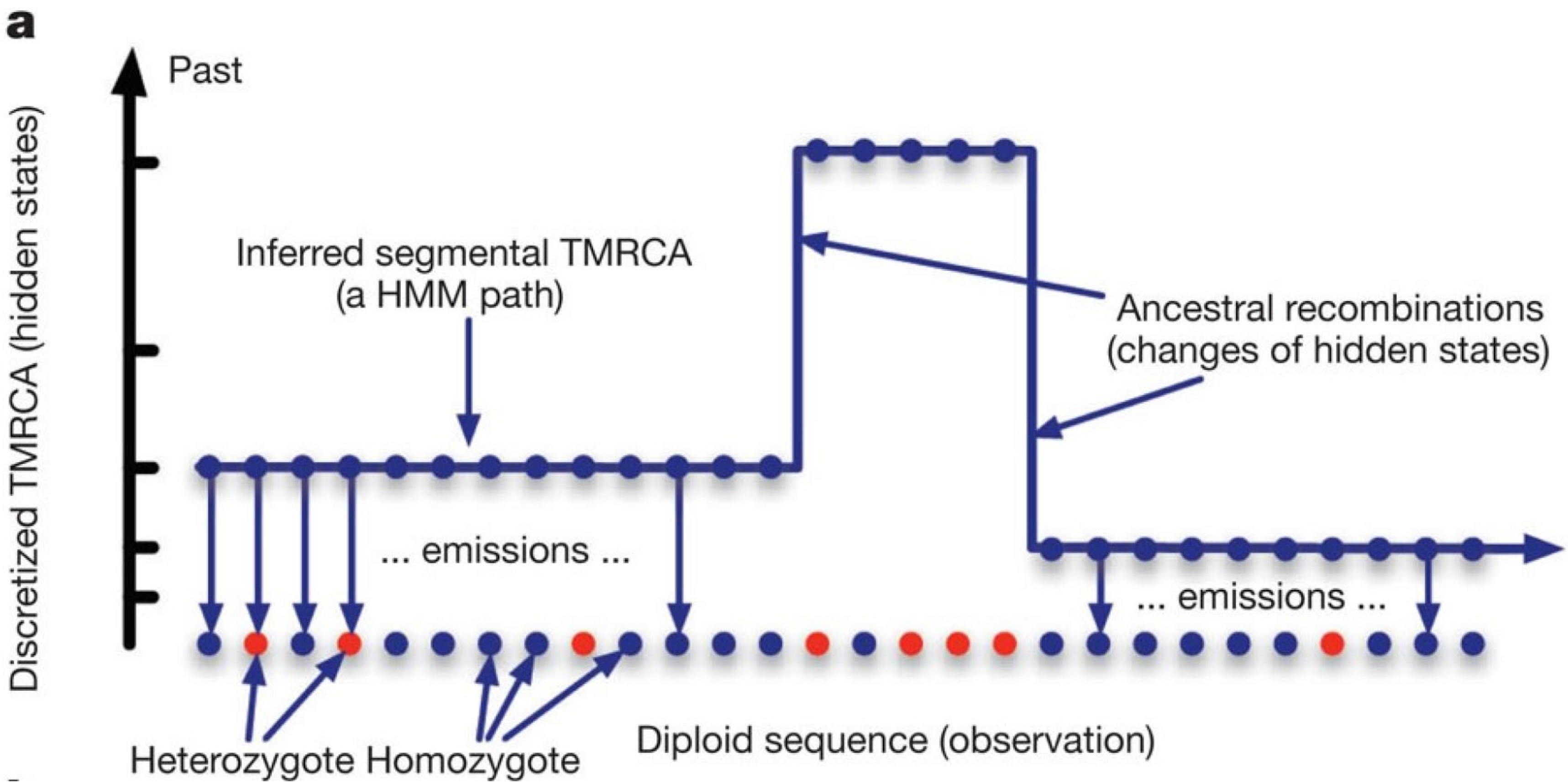
Nature Reviews | Genetics

PSMC – the basic principle

Segments of fixed TMRCA are separated by recombination

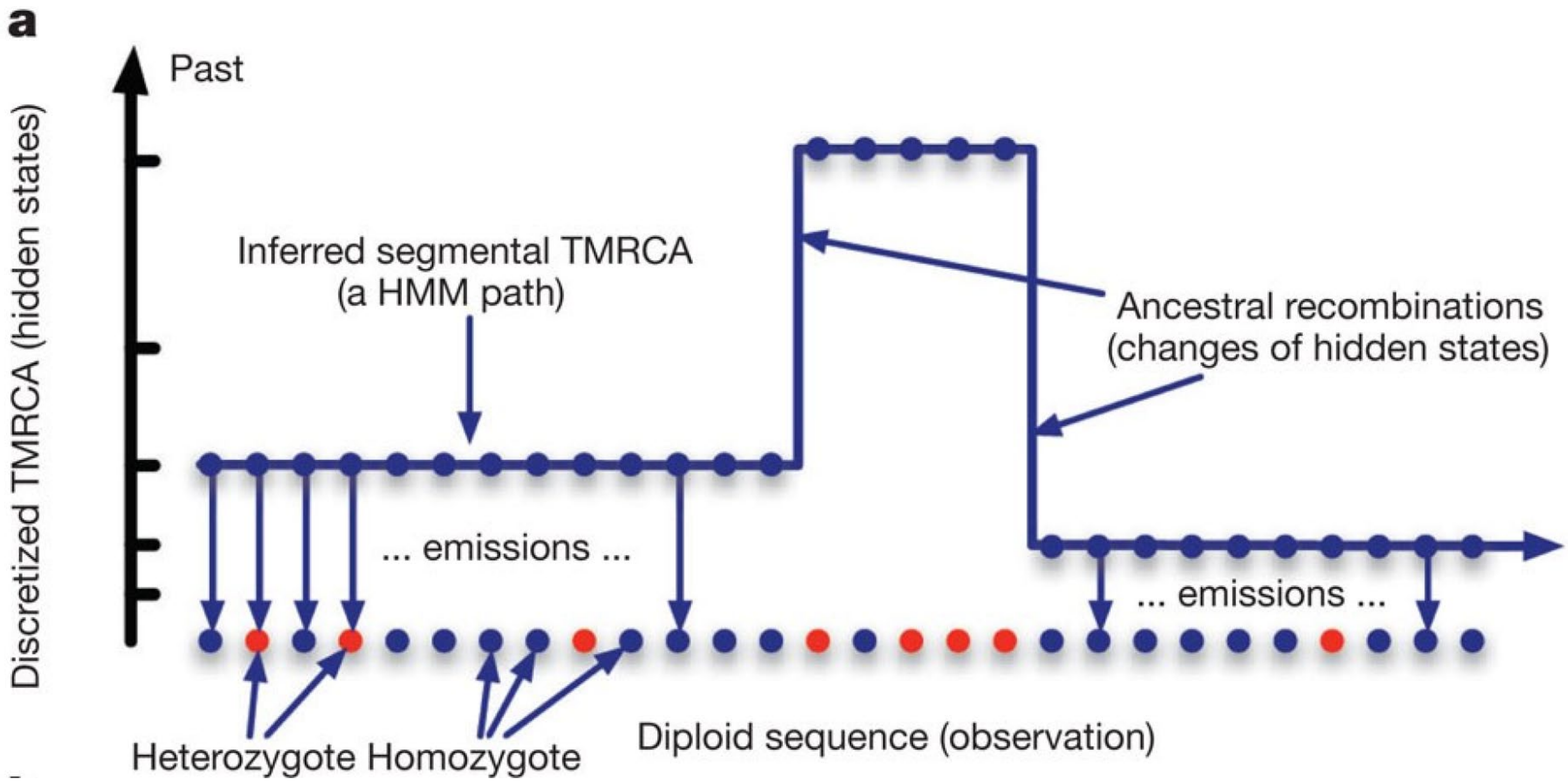


PSMC – a HMM



What is causing the model to estimate a change in TMRCA here?

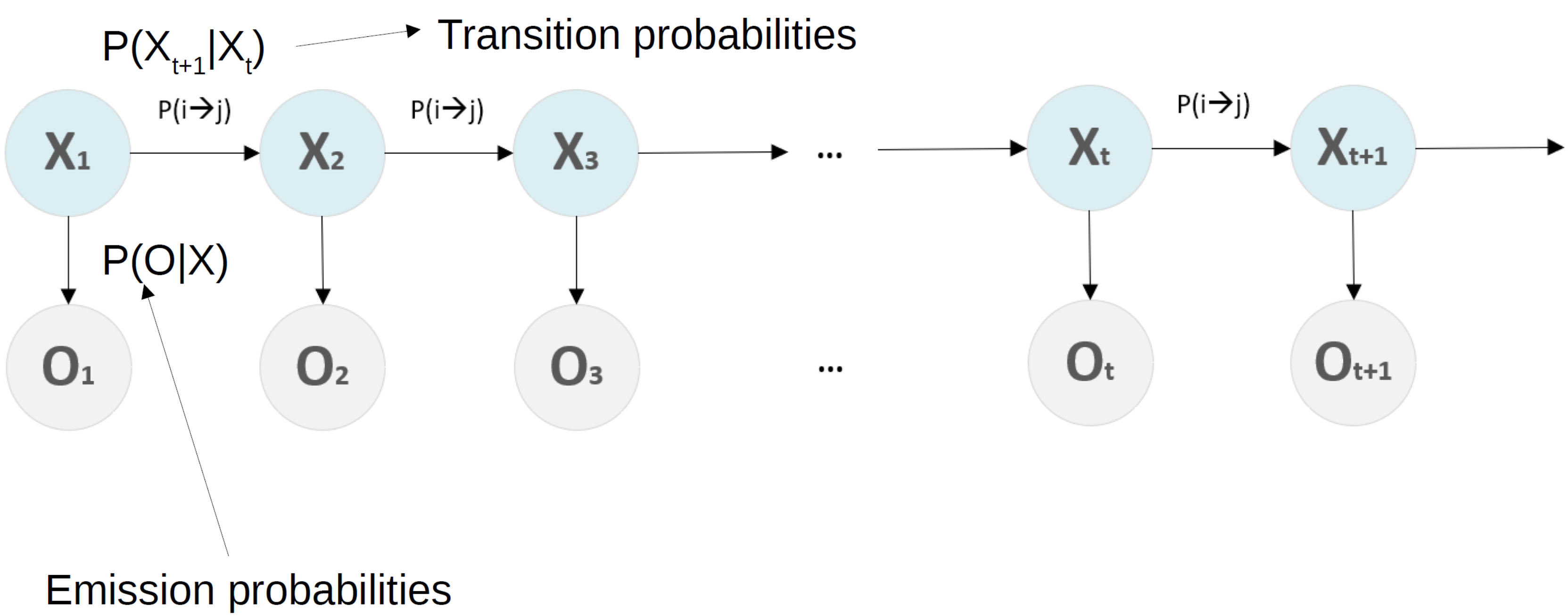
PSMC – a Hidden Markov Model

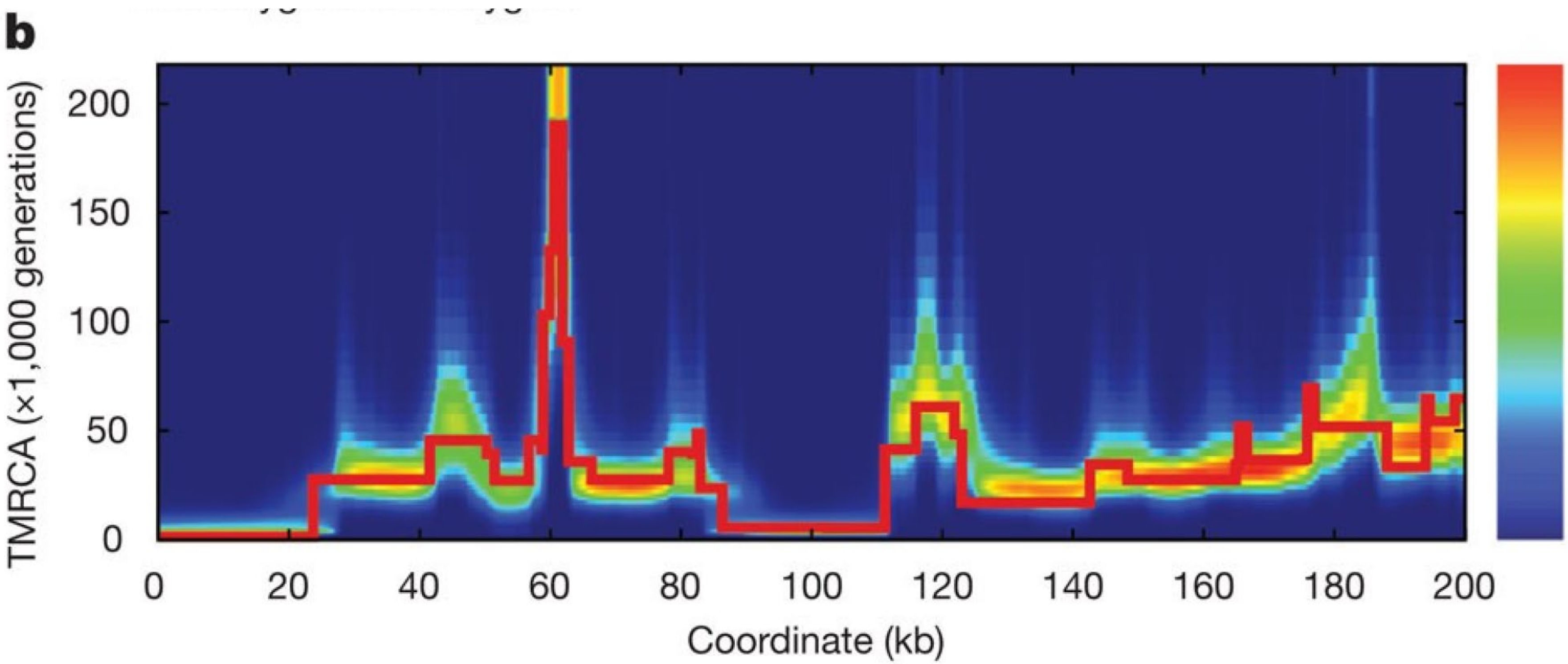


States: The possible values of TMRCA we will consider.
Emissions: The genotype data we observe.
Transitions: The changes from one TMRA to another, caused by recombination.

Components of a HMM

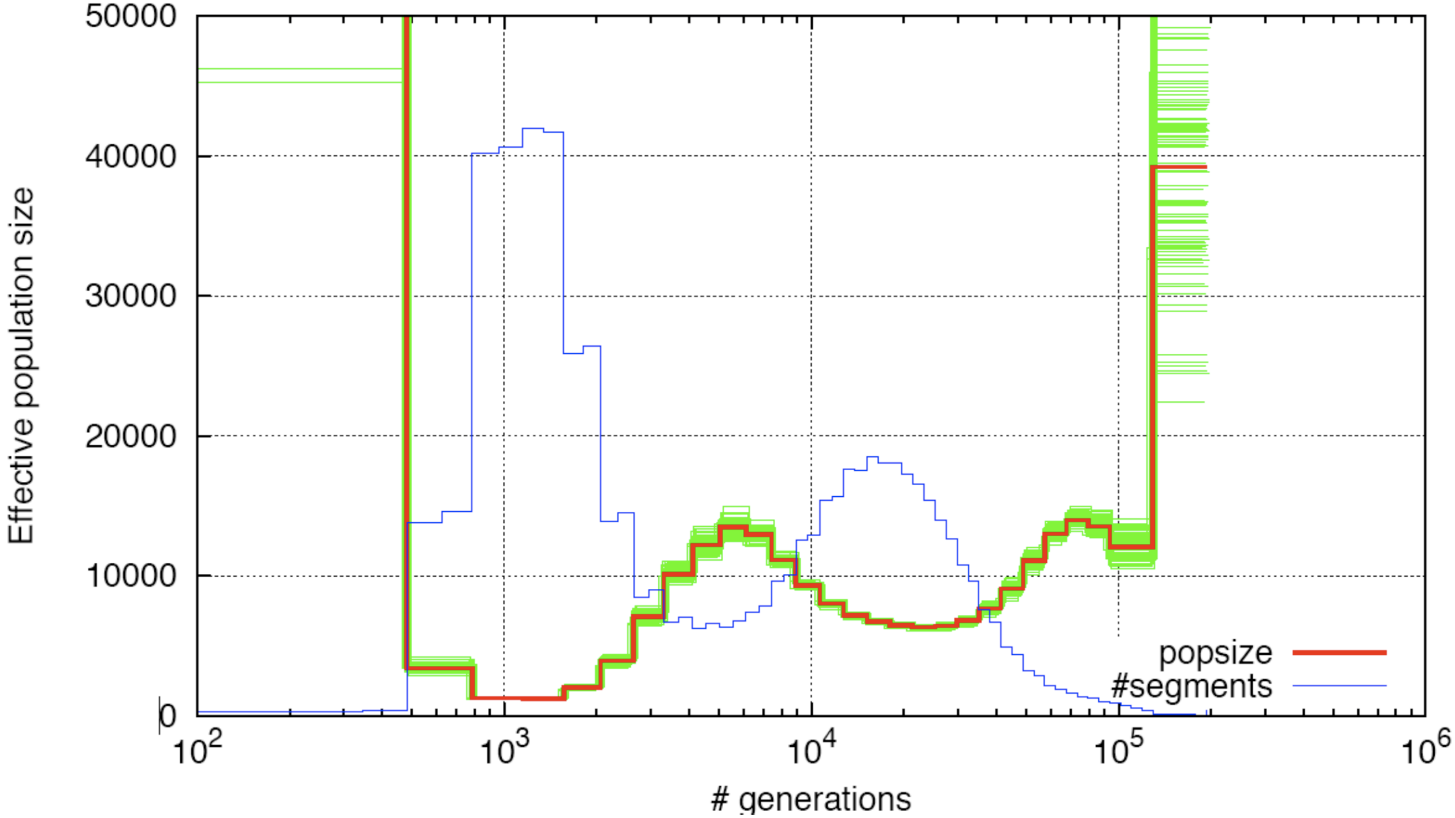
In this example there are two different **states**: i and j





PSMC mathematically translates the genome-wide distribution of TMRCAs into an estimate of N_E at different times.

Single human genome with bootstrap



Information is not equal in all time intervals

We can choose how we discretize time in PSMC

Why are human PSMCs different?